

LAPORAN DATA CLEANING

E-Commerce Sales Dataset — Day 1

Roadmap 7 Hari Data Analyst Portfolio — Dokumentasi Lengkap Langkah demi Langkah

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1. Pendahuluan

Laporan ini mendokumentasikan secara rinci setiap langkah proses pembersihan data (data cleaning) pada dataset transaksi e-commerce (raw_sales_data.csv) menggunakan Microsoft Excel, sebagai bagian dari Day 1 dalam roadmap pengembangan portofolio Data Analyst. Dataset mentah berisi 4.100 baris transaksi periode Januari 2023 – Desember 2024, dengan sejumlah masalah kualitas data yang sengaja merepresentasikan kondisi data dunia nyata: duplikasi, nilai kosong, inkonsistensi format, kesalahan tipe data, dan outlier.

Tujuan tahap ini adalah menghasilkan dataset yang bersih, konsisten, dan siap dipakai pada tahap selanjutnya (analisis Excel, SQL, Python, dan dashboard Tableau), sehingga angka yang dilaporkan pada seluruh tahap roadmap konsisten satu sama lain. Setiap langkah di bawah ini disertai formula yang dipakai dan tangkapan layar (screenshot) sebagai bukti proses, bukan hanya ringkasan naratif, sehingga dapat dijadikan referensi ulang tanpa perlu mengingat dari nol.

2. Ringkasan Eksekutif

Tabel 1 merangkum perubahan kondisi dataset sebelum dan sesudah proses cleaning.

Metrik	Sebelum	Sesudah
Jumlah baris	4.100	3.933
Jumlah kolom	11	12 (+ revenue)
Baris duplikat	99	0
Missing value (seluruh kolom)	533 sel	0
Outlier ekstrem (quantity & harga)	27 baris	0
Format order_date	5 format campur	1 format (YYYY-MM-DD)

Tabel 1. Perbandingan kondisi dataset sebelum dan sesudah cleaning

3. Tahapan Data Cleaning

3.1 Penghapusan Data Duplikat

Baris duplikat diidentifikasi menggunakan fitur Data > Remove Duplicates dengan seluruh kolom dicentang, sehingga hanya baris yang identik persis di semua kolom (termasuk order_id) yang dianggap duplikat — bukan sekadar kemiripan sebagian data. Pendekatan ini menghindari kesalahan menghapus dua transaksi berbeda yang kebetulan memiliki produk atau tanggal yang sama.

Langkah 1 — Identifikasi baris duplikat dengan row signature

Untuk memastikan hanya baris yang benar-benar identik yang tersorot, seluruh kolom digabung menjadi satu "tanda tangan baris" (row signature) memakai TEXTJOIN, lalu tanda tangan yang identik disorot dengan Conditional Formatting > Highlight Duplicate Values:

```
=TEXTJOIN("|",TRUE,A2:K2)
```

	D	E	F	G	H	I	J	K	L
	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	row_signature
1	Andi Halim	South	Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card	ORD100016 45253 CUST11353 Andi Halim Sports & Outdoors Water Bottle 2 28.97 Completed Credit Card
2	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	ORD100794 14899 CUST11478 Lisa Kurniawan East Toys Building Blocks 2 46.3 Returned Credit Card
3	Ahmad Halim	Mid-West	Beauty	Moisturizer	1	35.09	Completed	Gift Card	ORD103490 45392 CUST15312 Ahmad Halim Mid-West Beauty Moisturizer 1 35.09 Completed Gift Card
4	Ayu Hidayat	SOUTH	Sports & Outdoors	Camping Tent	2	12.94	Completed	Debit Card	ORD101567 45526 CUST1390 Ayu Hidayat SOUTH Sports & Outdoors Camping Tent 2 12.94 Completed Debit Card
5	James Yusuf	SOUTH	Sports & Outdoors	Hiking Backpack	1	23.82	Completed	PayPal	ORD100896 45348 CUST15463 James Yusuf SOUTH Sports & Outdoors Hiking Backpack 1 23.82 Completed PayPal
6	Fajar Halim	Midwest	Beauty	Sunscreen SPF50	1	33.88	Completed	Gift Card	ORD101473 45522 CUST1052 Fajar Halim Midwest Beauty Sunscreen SPF50 1 33.88 Completed Gift Card
7	John Tan	South	Books	Fiction Novel	2	10.88	Completed	Gift Card	ORD103111 45519 CUST1726 John Tan South Books Fiction Novel 2 10.88 Completed Gift Card
8	Kevin Anderson	WEST	Beauty	Moisturizer	1	21.46	Completed	PayPal	ORD100537 45119 CUST1445 Kevin Anderson WEST Beauty Moisturizer 1 21.46 Completed PayPal
9	John Yusuf	MIDWEST	Home & Kitchen	Cutting Board	1	102.68	Completed	PayPal	ORD100433 45040 CUST1699 John Yusuf MIDWEST Home & Kitchen Cutting Board 1 102.68 Completed PayPal
10	Ayu Smith	South	Sports & Outdoors	Resistance Bands	1	56.4	Completed	PayPal	ORD103151 45149 CUST1708 Ayu Smith South Sports & Outdoors Resistance Bands 1 56.4 Completed PayPal
11	Nina Lee	Midwest	Books	Business Strategy	1	18.34	Completed	Credit Card	ORD102767 45191 CUST1868 Nina Lee Midwest Books Business Strategy Book 1 18.34 Completed Credit Card
12	Rina Taylor	South	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	ORD101388 45366 CUST1211 South Clothing Winter Jacket 2 35.68 Completed Gift Card
13	Fajar Santos	East	Home & Kitchen	Storage Containers	1	112.68	Completed	Gift Card	ORD100780 45500 CUST1336 Fajar Santos East Home & Kitchen Storage Containers 1 112.68 Completed Gift Card
14	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	ORD101857 45143 CUST1104 Sarah Tan East Electronics Webcam 1 68.92 Completed Debit Card
15	Lisa Taylor	MIDWEST	Beauty	Shampoo	2	23.18	Cancelled		ORD100800 45314 CUST1352 Lisa Taylor MIDWEST Beauty Shampoo 2 23.18 Cancelled
16	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	ORD101695 45432 CUST1225 Budi Kurniawan West Clothing Hoodie 2 66.21 Returned PayPal
17	Linda Chen	East	Sports & Outdoors	Camping Tent	1		Cancelled	PayPal	ORD102093 45316 CUST1824 Linda Chen East Sports & Outdoors Camping Tent 1 Cancelled PayPal
18	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	ORD101296 45571 CUST1003 Siti Anderson East Electronics Power Bank 3 134.85 Completed PayPal
19	Siti Wilson	Midwest	Sports & Outdoors	Yoga Mat	1	75.34	Completed	Credit Card	ORD101842 45386 CUST1741 Siti Wilson Midwest Sports & Outdoors Yoga Mat 1 75.34 Completed Credit Card
20	Nina Garcia	West	Sports & Outdoors	Hiking Backpack	1	49.79	Returned	Credit Card	ORD102591 45223 CUST1856 Nina Garcia West Sports & Outdoors Hiking Backpack 1 49.79 Returned Credit Card
21	Robert Smith	West	Clothing	Denim Jeans	4	62.83	Completed	Debit Card	ORD101294 45157 CUST1210 Nina Garcia West Sports & Outdoors Hiking Backpack 1 49.79 Returned Credit Card
22	Siti Tan	EAST	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	ORD103824 45495 CUST1324 Robert Smith West Clothing Denim Jeans 4 62.83 Completed Debit Card
23	David Wijaya	EAST	Home & Kitchen	Blender	4	115.34	Completed	Debit Card	ORD101677 45654 CUST1227 David Wijaya EAST Home & Kitchen Blender 4 115.34 Completed Debit Card
24	David Anderson	South	Sports & Outdoors	Camping Tent	1	30.3	Completed	Debit Card	ORD102224 45051 CUST1588 David Anderson South Sports & Outdoors Camping Tent 1 30.3 Completed Debit Card
25	Sarah Johnson	South	Clothing	Hoodie	1	46.19	Completed	Debit Card	ORD101815 45477 CUST1265 Sarah Johnson South Clothing Hoodie 1 46.19 Completed Debit Card
26	Dewi Sari	EAST	Books	Self-Help Guide	1	29.66	Completed	Credit Card	ORD101819 44846 CUST1343 Dewi Sari EAST Books Self-Help Guide 1 29.66 Completed Credit Card
27	Putri Halim	East	SPORTS & Outdoors	Camping Tent	1	72.43	Cancelled	PayPal	ORD100756 45124 CUST1680 Putri Halim East SPORTS & OUTDOORS Camping Tent 1 72.43 Cancelled PayPal
28	Maria Wilson	SOUTH	TOYS	Board Game	1	58.15	Completed	PayPal	ORD101910 45024 CUST1887 Maria Wilson SOUTH TOYS Board Game 1 58.15 Completed PayPal
29	Dewi Tan	East	Beauty	Lip Balm	2	7.03	Completed	PayPal	ORD100326 45480 CUST1182 Dewi Tan East Beauty Lip Balm 2 7.03 Completed PayPal
30	Nina Garcia	MIDWEST	TOYS	RC Car	1	5.57	Completed		ORD103238 45032 CUST1254 Nina Garcia MIDWEST TOYS RC Car 1 5.57 Completed
31	Sarah Lee	East	Clothing	Cotton T-Shirt	1	54.55	Completed	Debit Card	ORD102634 45070 CUST1757 Sarah Lee East Clothing Cotton T-Shirt 1 54.55 Completed Debit Card
32	Rina Taylor	SOUTH	Beauty	Moisturizer	1	25.47	Completed	Debit Card	ORD103616 45348 CUST1128 Rina Taylor SOUTH Beauty Moisturizer 1 25.47 Completed Debit Card
33	Ahmad Hidayat	Mid-West	Home & Kitchen	Coffee Maker	2	52.50	Cancelled	Gift Card	ORD101919 45245 CUST1308 Ahmad Hidayat Mid-West Home & Kitchen Coffee Maker 2 52.50 Cancelled Gift Card
34	Siti Wilson	WEST	Beauty	Shampoo	2	544.8	Completed	PayPal	ORD103226 45042 CUST1206 Siti Wilson WEST Beauty Shampoo 2 544.8 Completed PayPal
35	Sarah Chen	WEST	Toys	Board Game	1	58.26	Completed	Debit Card	ORD102507 45192 CUST1643 Sarah Chen WEST Toys Board Game 1 58.26 Completed Debit Card
36	Rina Anderson	WEST	Toys	Puzzle 1000pc	1	25.68	Completed	Debit Card	ORD100687 44943 CUST1370 Rina Anderson WEST Toys Puzzle 1000pc 1 25.68 Completed Debit Card
37	Rudi Chen	South	ELECTRONICS	Bluetooth Speaker	2	93.44	Completed	PayPal	ORD102423 45047 CUST1838 Rudi Chen South ELECTRONICS Bluetooth Speaker 2 93.44 Completed PayPal
38	Kevin Lee	East	Clothing	Running Shorts	2	24.84	Completed	PayPal	ORD100333 45560 CUST1753 Kevin Lee East Clothing Running Shorts 2 24.84 Completed PayPal
39	Putri Anderson	West	Clothing	Hoodie	2		Completed	PayPal	ORD102841 17/09/2023 CUST1635 Putri Anderson West Clothing Hoodie 2 Completed PayPal
40	Nina Garcia	SOUTH	Beauty	Moisturizer	2	104.8	Completed	Gift Card	ORD100687 44893 CUST1887 Nina Garcia SOUTH Beauty Moisturizer 2 104.8 Completed Gift Card
41	Ayu Chen	South	ELECTRONIC	Power Bank	2	142.03	Completed	Debit Card	ORD100619 45464 CUST1454 Ayu Chen South ELECTRONICS Power Bank 2 142.03 Completed Debit Card
42	Dewi Lee	Midwest	Beauty	Sunscreen SPF50	2	15.43	Completed	Debit Card	ORD101200 45232 CUST1608 Dewi Lee Midwest Beauty Sunscreen SPF50 2 15.43 Completed Debit Card
43	Maria Anderson	South	TOYS	Puzzle 1000pc	1	19.88	Completed	Debit Card	ORD102075 45581 CUST1794 Maria Anderson South TOYS Puzzle 1000pc 1 19.88 Completed Debit Card
44	Ayu Pratama	East	Sports & Outdoors	Yoga Mat	1	38.17	Completed	PayPal	ORD101211 24/03/2024 CUST1073 Ayu Pratama East Sports & Outdoors Yoga Mat 1 38.17 Completed PayPal
45	Sarah Tan	MIDWEST	Clothing	Cotton T-Shirt	4	38.53	Returned		ORD102968 44883 CUST1383 Sarah Tan MIDWEST Clothing Cotton T-Shirt 4 38.53 Returned

Gambar 1. Kolom row_signature — baris dengan tanda tangan identik tersorot warna, menandakan duplikat asli

Langkah 2 — Remove Duplicates

Setelah baris duplikat terverifikasi, dijalankan Data > Remove Duplicates dengan seluruh kolom dicentang:

Gambar 2. Hasil Remove Duplicates — 98 baris duplikat ditemukan dan dihapus, menyisakan 4.002 baris unik

- Hasil awal: 98 baris exact-duplicate dihapus, menyisakan 4.002 baris unik.

Langkah 3 — Verifikasi ulang (Day 2)

Sebelum dataset dipakai untuk tahap analisis, dilakukan pengecekan ulang dengan teknik yang sama (row signature + filter warna). Ditemukan 1 baris duplikat tambahan (order_id ORD101170, customer Fajar Wijaya — Wool Socks, qty 3) yang lolos dari proses awal:

Gambar 3. Verifikasi ulang di Day 2 menemukan 1 duplikat tambahan yang lolos dari pembersihan awal (baris 441 dan 3935, isi identik)

- Total akhir: 99 baris exact-duplicate dihapus (98 pada pembersihan awal + 1 ditemukan saat verifikasi ulang).

3.2 Standardisasi Kolom Category & Region

Kedua kolom ini memiliki variasi kapitalisasi ("electronics", "ELECTRONICS"), spasi berlebih, dan kesalahan penulisan ("Cloths", "Electronic", "Mid-West"). Standardisasi dilakukan melalui kolom bantu dengan formula `PROPER + TRIM`, dilanjutkan koreksi typo manual via Find & Replace, lalu hasilnya di-paste kembali (Paste Special > Values) ke kolom asli:

```
=PROPER(TRIM([@category]))
```

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	category_clean	
1	ORD100016	11/23/2023	CUST1353	Andi Halim		Sports & Ou	Water Bottle	2	28.97	Completed	Credit Card	Sports & Outdoors	
2	ORD102704	3/14/2023	CUST1478	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	F3))	
4	ORD103490	4/10/2024	CUST1531	Ahmad Halim	Mid-West	Beauty	Moisturizer	1	35.09	Completed	Gift Card	Beauty	
5	ORD101567	8/22/2024	CUST1390	Ayu Hidayat	SOUTH	Sports & Ou	Camping Tent	2	12.94	Completed	Debit Card	Sports & Outdoors	
6	ORD101096	21-Feb-24	CUST1563	James Yusuf	SOUTH	sports & ou	Hiking Backpack	1	23.12	Completed	PayPal	Sports & Outdoors	
7	ORD101473	8/18/2024	CUST1052	Fajar Halim	Midwest	Beauty	Sunscreen SPF50	1	33.88	Completed	Gift Card	Beauty	
8	ORD103111	8/15/2024	CUST1726	John Tan	south	books	Fiction Novel	2	10.88	Completed	Gift Card	Books	
9	ORD100537	1/28/2023	CUST1445	Kevin Anderson	WEST	Beauty	Moisturizer	1	21.46	Completed	PayPal	Beauty	
10	ORD102413	24-Apr-23	CUST1699	John Yusuf	MIDWEST	Home & Kit	Cutting Board	1	102.66	Completed	PayPal	Home & Kitchen	
11	ORD103151	11-Aug-23	CUST1708	Ayu Smith	South	sports & ou	Resistance Bands	1	56.4	Completed	PayPal	Sports & Outdoors	
12	ORD102767	9/23/2023	CUST1868	Nina Lee	midwest	Books	Business Strategy	1	18.14	Completed	Credit Card	Books	
13	ORD101868	15-Mar-24	CUST1211		south	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	Clothing	
14	ORD100749	7/27/2024	CUST1334	Fajar Santos	east	Home & Kit	Storage Container	1	112.68	Completed	Gift Card	Home & Kitchen	
15	ORD101857	8/5/2023	CUST1104	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	Electronics	
16	ORD100800	23-Jan-24	CUST1352	Lisa Taylor	MIDWEST	Beauty	Shampoo	2	23.18	Cancelled		Beauty	
17	ORD101695	20-May-24	CUST1255	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	Clothing	
18	ORD102093	1/25/2024	CUST1824	Linda Chen	east	Sports & Ou	Camping Tent	1		Cancelled	PayPal	Sports & Outdoors	
19	ORD101296	10/6/2024	CUST1003	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	Electronics	
20	ORD103842	9/17/2023	CUST1741	Siti Wilson	midwest	Sports & Ou	Yoga Mat	1	75.14	Completed	Credit Card	Sports & Outdoors	
21	ORD102931	10/26/2023	CUST1856		SOUTH	electronics	Power Bank	5	126.65	Completed	Gift Card	Electronics	
22	ORD102954	8/19/2023	CUST1210	Nina Garcia	West	Sports & Ou	Hiking Backpack	1	49.73	Returned	Credit Card	Sports & Outdoors	
23	ORD103824	7/22/2024	CUST1524	Robert Smith	west	clothing	Denim Jeans	4	62.83	Completed	Debit Card	Clothing	
24	ORD101074	4/12/2024	CUST1689	Siti Tan	EAST	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	Beauty	

Gambar 4. Kolom bantu category_clean memakai formula PROPER(TRIM()))

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	region_trim	
1	ORD100016	2023-11-23	CUST1353	Andi Halim		Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card		
2	ORD102704	2023-03-14	CUST1478	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	East	
3	ORD103490	2024-04-10	CUST1531	Ahmad Halim	Midwest	Beauty	Moisturizer	1	35.09	Completed	Gift Card	Midwest	
4	ORD101567	2024-08-22	CUST1390	Ayu Hidayat	South	Sports & Outdoors	Camping Tent	2	12.94	Completed	Debit Card	South	
5	ORD101096	2024-02-21	CUST1563	James Yusuf	South	Sports & Outdoors	Hiking Backpack	1	23.12	Completed	PayPal	South	
6	ORD101473	2024-08-18	CUST1052	Fajar Halim	Midwest	Beauty	Sunscreen SPF50	1	33.88	Completed	Gift Card	Midwest	
7	ORD103111	2024-08-15	CUST1726	John Tan	South	Books	Fiction Novel	2	10.88	Completed	Gift Card	South	
8	ORD100537	2024-01-28	CUST1445	Kevin Anderson	West	Beauty	Moisturizer	1	21.46	Completed	PayPal	West	
9	ORD102413	2023-04-24	CUST1699	John Yusuf	Midwest	Home & Kitchen	Cutting Board	1	102.66	Completed	PayPal	Midwest	
10	ORD103151	2023-08-11	CUST1708	Ayu Smith	South	Sports & Outdoors	Resistance Bands	1	56.4	Completed	PayPal	South	
11	ORD102767	2023-09-22	CUST1868	Nina Lee	Midwest	Books	Business Strategy	1	18.14	Completed	Credit Card	Midwest	
12	ORD101868	2024-03-15	CUST1211		South	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	South	
13	ORD100749	2024-07-27	CUST1334	Fajar Santos	East	Home & Kitchen	Storage Container	1	112.68	Completed	Gift Card	East	
14	ORD101857	2023-08-05	CUST1104	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	East	
15	ORD100800	2024-02-23	CUST1352	Lisa Taylor	Midwest	Beauty	Shampoo	2	23.18	Cancelled		Midwest	
16	ORD101695	2024-05-20	CUST1255	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	West	
17	ORD102093	2024-01-25	CUST1824	Linda Chen	East	Sports & Outdoors	Camping Tent	1		Cancelled	PayPal	East	
18	ORD101296	2024-10-06	CUST1003	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	East	
19	ORD103842	2023-09-17	CUST1741	Siti Wilson	Midwest	Sports & Outdoors	Yoga Mat	1	75.14	Completed	Credit Card	Midwest	
20	ORD102931	2023-10-26	CUST1856		South	Electronics	Power Bank	5	126.65	Completed	Gift Card	South	
21	ORD102954	2023-08-19	CUST1210	Nina Garcia	West	Sports & Outdoors	Hiking Backpack	1	49.73	Returned	Credit Card	West	
22	ORD103824	2024-07-22	CUST1524	Robert Smith	West	Clothing	Denim Jeans	4	62.83	Completed	Debit Card	West	
23	ORD101074	2024-04-12	CUST1689	Siti Tan	East	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	East	
24	ORD101777	2024-12-28	CUST1227	David Wijaya	East	Home & Kitchen	Blender	4	115.54	Completed	Debit Card	East	
25	ORD102224	2023-05-05	CUST1588	David Anderson	South	Sports & Outdoors	Camping Tent	1	30.3	Completed	Debit Card	South	
26	ORD103815	2024-07-04	CUST1265	Sarah Johnson	South	Clothing	Hoodie	1	46.19	Completed	Debit Card	South	
27	ORD101819	2023-02-09	CUST1343	Dewi Sari	East	Books	Self-Help Guide	1	25.66	Completed	Credit Card	East	
28	ORD100756	2023-07-17	CUST1680	Putri Halim	East	Sports & Outdoors	Camping Tent	1	72.43	Cancelled	PayPal	East	
29	ORD101910	2023-04-08	CUST1987	Maria Wilton	South	Toys	Board Game	1	58.15	Completed	PayPal	South	
30	ORD100326	2024-07-07	CUST1182	Dewi Tan	East	Beauty	Lip Balm	2	7.83	Completed	PayPal	East	
31	ORD103238	2023-04-16	CUST1254	Nina Garcia	Midwest	Toys	RC Car	1	9.57	Completed		Midwest	
32	ORD102634	2023-05-24	CUST1757	Sarah Lee	East	Clothing	Cotton T-Shirt	1	54.55	Completed	Debit Card	East	
33	ORD103616	2024-02-26	CUST1128	Rina Taylor	South	Beauty	Moisturizer	1	25.47	Completed	Debit Card	South	
34	ORD101939	2023-11-15	CUST1806	Ahmad Hidayat	Midwest	Home & Kitchen	Coffee Maker	2	52.95	Cancelled	Gift Card	Midwest	
35	ORD103226	2024-09-07	CUST1206	Siti Wilson	West	Beauty	Shampoo	2	\$44.80	Completed	PayPal	West	
36	ORD102507	2023-09-23	CUST1645	Sarah Chen	West	Toys	Board Game	1	58.26	Completed	Debit Card	West	
37	ORD100857	2023-01-17	CUST1370	Rina Anderson	West	Toys	Puzzle 1000pc	1	25.68	Completed	Debit Card	West	
38	ORD102423	2023-05-01	CUST1838	Budi Chen	South	Electronics	Bluetooth Speaker	2	93.44	Completed	PayPal	South	
39	ORD100333	2024-09-25	CUST1753	Kevin Lee	East	Clothing	Running Shorts	2	24.84	Completed	PayPal	East	
40	ORD102984	2023-09-17	CUST1615	Putri Anderson	West	Clothing	Hoodie	2		Completed	PayPal	West	
41	ORD100087	2023-03-06	CUST1867	Nina Garcia	South	Beauty	Moisturizer	2	10.48	Completed	Gift Card	South	
42	ORD100619	2024-06-21	CUST1414	Ayu Chen	South	Electronics	Power Bank	2	142.03	Completed	Debit Card	South	

Gambar 5. Kolom category dan region setelah distandarisasi (Proper Case, tanpa typo)

3.3 Standardisasi Format Tanggal (order_date)

Kolom order_date awalnya berisi 5 format berbeda (YYYY-MM-DD, MM/DD/YYYY, DD-Mon-YYYY, DD/MM/YYYY, YYYY/MM/DD). Baris yang masih tersimpan sebagai teks diidentifikasi terlebih dahulu dengan ISTEEXT:

```
=ISTEXT([@order_date])
```

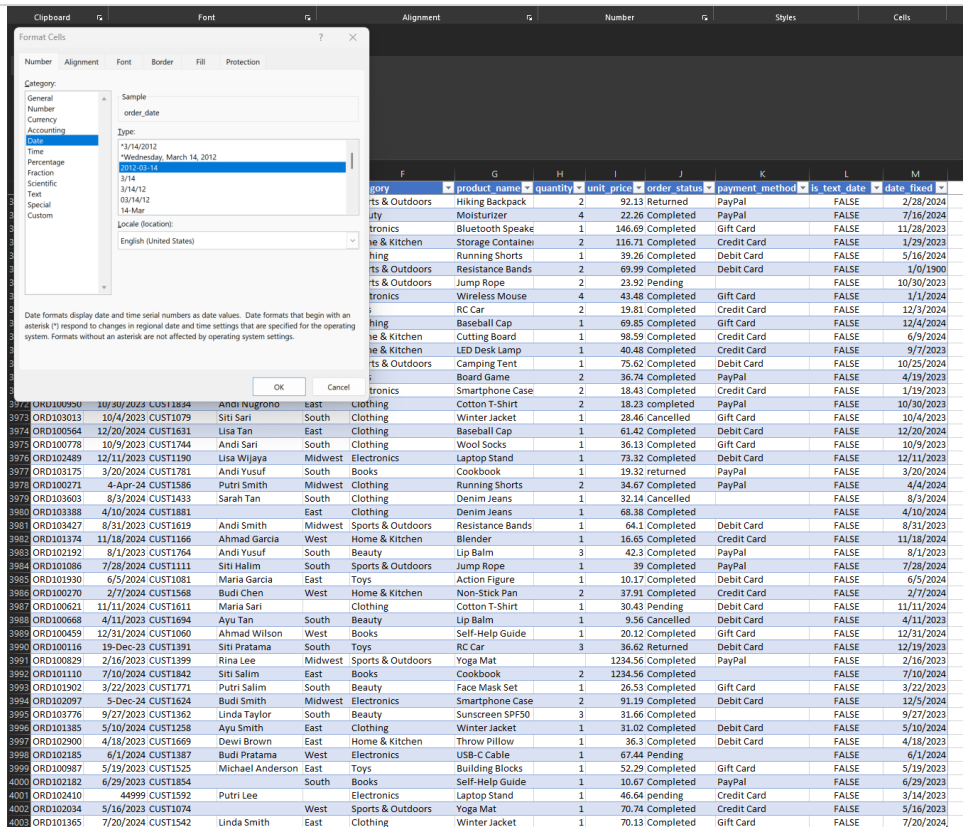
	id	is_text_date
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
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		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		FALSE
		TRUE
		FALSE
		FALSE

Gambar 6. Kolom bantu `is_text_date — TRUE` menandakan baris yang `order_date`-nya masih tersimpan sebagai teks

Baris yang bernilai TRUE dikonversi menggunakan formula berikut, yang membaca format DD/MM/YYYY secara eksplisit dan membiarkan tanggal yang sudah valid (FALSE) tidak berubah:

```
=IF(ISTEXT([@order_date]), DATE(VALUE(RIGHT([@order_date],4)),  
VALUE(MID([@order_date],4,2)), VALUE(LEFT([@order_date],2))), [@order_date])
```

Setelah seluruh nilai berhasil dikonversi menjadi tanggal asli, format tampilan diseragamkan lewat Format Cells > Date > custom type YYYY-MM-DD (bukan format beraster yang mengikuti regional setting Windows):



Gambar 7. Dialog Format Cells — memilih format tanggal YYYY-MM-DD yang konsisten di semua sistem

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	is_text_date	date_fixed
1	ORD100016	2023-11-23	CUST1353	Andi Halim	East	Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card	FALSE	2/28/2024
2	ORD102704	2023-03-14	CUST1478	Lisa Kurniawan	East	Toys	Building Blocks	2	46.3	Returned	Credit Card	FALSE	7/16/2024
3	ORD103490	2024-04-10	CUST1531	Ahmad Halim	Midwest	Beauty	Moisturizer	1	35.09	Completed	Gift Card	FALSE	11/28/2023
4	ORD101567	2024-08-22	CUST1390	Ayu Hidayat	SOUTH	Sports & Outdoors	Camping Tent	2	12.94	Completed	Debit Card	FALSE	1/29/2023
5	ORD101096	2024-02-21	CUST1563	James Yusuf	SOUTH	Sports & Outdoors	Hiking Backpack	1	23.12	Completed	PayPal	FALSE	5/16/2024
6	ORD101473	2024-08-18	CUST1052	Fajar Halim	Midwest	Beauty	Sunscreens SPF50	1	33.88	Completed	Gift Card	FALSE	1/0/1900
7	ORD103111	2024-08-15	CUST1726	John Tan	South	Books	Fiction Novel	2	10.88	Completed	Gift Card	FALSE	10/30/2023
8	ORD100537	2024-01-28	CUST1445	Kevin Anderson	WEST	Beauty	Moisturizer	1	21.46	Completed	PayPal	FALSE	1/1/2024
9	ORD102413	2023-04-24	CUST1699	John Yusuf	MIDWEST	Home & Kitchen	Cutting Board	1	102.66	Completed	PayPal	FALSE	12/3/2024
10	ORD103151	2023-08-11	CUST1708	Ayu Smith	South	Sports & Outdoors	Resistance Bands	1	56.4	Completed	PayPal	FALSE	12/4/2024
11	ORD102767	2023-09-22	CUST1868	Nina Lee	Midwest	Books	Business Strategy	1	18.14	Completed	Credit Card	FALSE	6/9/2024
12	ORD101868	2024-03-15	CUST1211		South	Clothing	Winter Jacket	2	35.68	Completed	Gift Card	FALSE	9/7/2023
13	ORD100749	2024-07-27	CUST1334	Fajar Santos	East	Home & Kitchen	Storage Containers	1	112.68	Completed	Gift Card	FALSE	10/25/2024
14	ORD101857	2023-08-05	CUST1104	Sarah Tan	East	Electronics	Webcam	1	68.92	Completed	Debit Card	FALSE	4/19/2023
15	ORD100800	2024-01-23	CUST1352	Lisa Taylor	MIDWEST	Beauty	Shampoo	2	23.18	Cancelled		FALSE	10/25/2024
16	ORD101695	2024-05-20	CUST1255	Budi Kurniawan	West	Clothing	Hoodie	2	66.21	Returned	PayPal	FALSE	4/19/2023
17	ORD102093	2024-01-25	CUST1824	Linda Chen	East	Sports & Outdoors	Camping Tent	1		Cancelled	PayPal	FALSE	1/1/2024
18	ORD101296	2024-10-06	CUST1003	Siti Anderson	East	Electronics	Power Bank	3	134.85	Completed	PayPal	FALSE	12/3/2024
19	ORD103842	2023-09-17	CUST1741	Siti Wilson	Midwest	Sports & Outdoors	Yoga Mat	1	75.14	Completed	Credit Card	FALSE	12/4/2024
20	ORD102931	2023-10-26	CUST1856		SOUTH	Electronics	Power Bank	5	126.65	Completed	Gift Card	FALSE	6/9/2024
21	ORD102954	2023-08-19	CUST1210	Nina Garcia	West	Sports & Outdoors	Hiking Backpack	1	49.73	Returned	Credit Card	FALSE	9/7/2023
22	ORD103824	2024-07-22	CUST1524	Robert Smith	West	Clothing	Denim Jeans	4	62.83	Completed	Debit Card	FALSE	12/2/2024
23	ORD101074	2024-04-12	CUST1689	Siti Tan	EAST	Beauty	Face Mask Set	1	39.05	Completed	Debit Card	FALSE	12/12/2023
24	ORD101777	2024-12-28	CUST1227	David Wijaya	EAST	Home & Kitchen	Blender	4	115.54	Completed	Debit Card	FALSE	10/9/2023
25	ORD102224	2023-05-05	CUST1588	David Anderson	South	Sports & Outdoors	Camping Tent	1	30.3	Completed	Debit Card	FALSE	12/12/2023
26	ORD103815	2024-07-04	CUST1265	Sarah Johnson	South	Clothing	Hoodie	1	46.19	Completed	Debit Card	FALSE	3/20/2024
27	ORD101819	2023-02-09	CUST1343	Dewi Sari	EAST	Books	Self-Help Guide	1	25.66	Completed	Credit Card	FALSE	2/7/2024
28	ORD100756	2023-07-17	CUST1680	Putri Halim	East	Sports & Outdoors	Camping Tent	1	72.43	Cancelled	PayPal	FALSE	11/1/2024
29	ORD101910	2023-04-08	CUST1887	Maria Wilson	SOUTH	Toys	Board Game	1	58.15	Completed	PayPal	FALSE	4/1/2023
30	ORD100326	2024-07-07	CUST1182	Dewi Tan	East	Beauty	Lip Balm	2	7.83	Completed	PayPal	FALSE	12/31/2024
31	ORD103238	2023-04-16	CUST1254	Nina Garcia	MIDWEST	Toys	RC Car	1	9.57	Completed		FALSE	4/1/2023
32	ORD102634	2023-05-24	CUST1757	Sarah Lee	East	Clothing	Cotton T-Shirt	1	54.55	Completed	Debit Card	FALSE	7/28/2024
33	ORD103616	2024-02-26	CUST1128	Rina Taylor	SOUTH	Beauty	Moisturizer	1	25.47	Completed	Debit Card	FALSE	8/1/2023
34	ORD101939	2023-11-15	CUST1806	Ahmad Hidayat	Midwest	Home & Kitchen	Coffee Maker	2	52.95	Cancelled	Gift Card	FALSE	8/1/2023
35	ORD103226	2024-09-07	CUST1206	Siti Wilson	WEST	Beauty	Shampoo	2	\$44.80	Completed	PayPal	FALSE	8/1/2023
36	ORD102507	2023-09-23	CUST1645	Sarah Chen	WEST	Toys	Board Game	1	58.26	Completed	Debit Card	FALSE	12/19/2023
37	ORD100857	2023-01-17	CUST1370	Rina Anderson	WEST	Toys	Puzzle 1000pc	1	25.68	Completed	Debit Card	FALSE	10/30/2023
38	ORD102423	2023-05-01	CUST1838	Budi Chen	South	Electronics	Bluetooth Speaker	2	93.44	Completed	PayPal	FALSE	10/4/2023
39	ORD100333	2024-09-25	CUST1753	Kevin Lee	East	Clothing	Running Shorts	2	24.84	Completed	PayPal	FALSE	12/12/2023
40	ORD102984	2023-09-17	CUST1615	Putri Anderson	West	Clothing	Hoodie	2		Completed	PayPal	FALSE	3/20/2024
41	ORD100087	2023-03-06	CUST1867	Nina Garcia	SOUTH	Beauty	Moisturizer	2	10.48	Completed	Gift Card	FALSE	5/16/2023
42	ORD100619	2024-06-21	CUST1414	Ayu Chen	South	Electronics	Power Bank	2	142.03	Completed	Debit Card	FALSE	7/20/2024

Gambar 8. Kolom order_date setelah seluruh nilai dikonversi dan diformat seragam

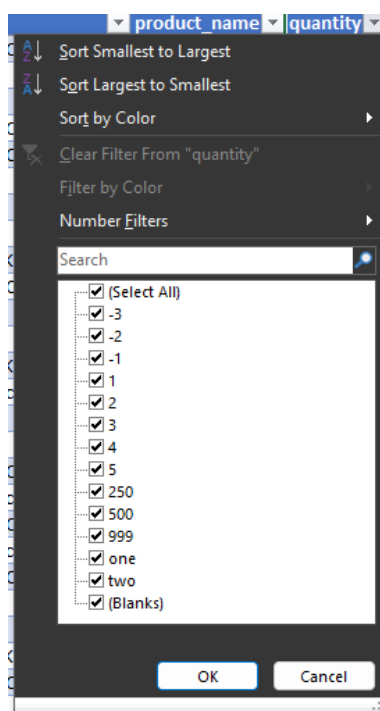
- Seluruh kolom order_date diformat ulang secara seragam ke YYYY-MM-DD.
- Satu nilai tanggal yang tersimpan sebagai 0 (bekas sel kosong) ditemukan lewat pengecekan `=MIN(order_date)` dan dikembalikan menjadi blank agar tidak menyamarkan missing value sebagai tanggal 1900.

	A	B	C	D	E	F
1	Hari	Fase	Task	Tools	Selesai?	
2	Day 1	Excel Data Cleaning	Ambil dataset e-commerce sales	Excel	SELESAI	
3	Day 1	Excel Data Cleaning	Cek & tangani missing values	Excel		
4	Day 1	Excel Data Cleaning	Cek & hapus duplicate records	Excel	SELESAI	
5	Day 1	Excel Data Cleaning	Perbaiki data types yang salah	Excel		
6	Day 1	Excel Data Cleaning	Standarkan kategori yang tidak konsisten	Excel	SELESAI	
7	Day 1	Excel Data Cleaning	Format ulang kolom tanggal	Excel	SELESAI	
8	Day 1	Excel Data Cleaning	Cek outlier	Excel		
9	Day 1	Excel Data Cleaning	Buat calculated columns	Excel		
10	Day 1	Excel Data Cleaning	Dokumentasikan raw -> cleaning -> clean dataset	Excel		
11	Day 1	Excel Data Cleaning	Simpan sebagai cleaned sales data.xlsx	Excel		

Gambar 9. Checklist Day 1 setelah tahap duplicate, data types, category/region, dan tanggal selesai

3.4 Perbaiki Tipe Data dan Outlier (Quantity & Unit Price)

Kolom quantity dan unit_price memiliki beberapa masalah sekaligus: nilai bertipe teks ("one", "two"), nilai negatif, harga tersimpan sebagai teks berformat "\$xx.xx", serta outlier ekstrem yang tidak realistis untuk transaksi retail individual.



Gambar 10. Variasi nilai bermasalah pada kolom quantity, dilihat lewat filter dropdown (negatif, teks, dan outlier ekstrem tercampur)

G	H	I	J	K	L
product_name	quantity	unit_price	order_status	payment_method	quantity_clean
Wool Socks	-3	64.63	Completed	PayPal	3
Fiction Novel	-3	30.39	Cancelled	Credit Card	3

Gambar 11. Contoh baris quantity bernilai negatif (-3) sebelum diperbaiki

Perbaiki tipe data quantity

```
quantity_clean:
=IFERROR(IF([@quantity]="one",1,IF([@quantity]="two",2,IF([@quantity]="three",3,ABS(VALUE([@quantity]))))),"
```

Perbaiki tipe data unit_price

	H	I	J	K	L
	quantity	unit_price	order_status	payment_method	quantity_clean
	2	\$44.80	Completed	PayPal	2
	1	\$75.09	Completed	PayPal	1
	1	\$86.87	Completed	PayPal	1
	1	\$69.18	Completed	Credit Card	1
	1	\$17.17	Completed	Debit Card	1
	1	\$60.06	Completed	Gift Card	1
	1	\$20.53	Completed	PayPal	1
	1	\$31.32	Completed	PayPal	1
	1	\$106.64	Completed	Debit Card	1
	2	\$34.82	Completed	Credit Card	2
	1	\$41.30	Completed	Gift Card	1
	1	\$130.75	Completed	Credit Card	1
	1	\$29.64	Completed		1
	1	\$16.59	completed	Debit Card	1
	1	\$38.05	Completed	Gift Card	1
	1	\$44.88	Completed	Credit Card	1
	2	\$12.32	Completed	Debit Card	2
	2	\$75.42	Completed	Debit Card	2
	3	\$15.48	cancelled	Gift Card	3

Gambar 12. Kolom unit_price masih berformat teks "\$xx.xx" sebelum dibersihkan

```
unit_price_clean:
=IFERROR(VALUE(SUBSTITUTE([@unit_price],"$","")), "")
```

	H	I	J	K	L	M
	quantity	unit_price	order_status	payment_method	quantity_clean	unit_price
	2	\$44.80	Completed	PayPal	2	44.8
	1	\$75.09	Completed	PayPal	1	75.09
	1	\$86.87	Completed	PayPal	1	86.87
	1	\$69.18	Completed	Credit Card	1	69.18
	1	\$17.17	Completed	Debit Card	1	17.17
	1	\$60.06	Completed	Gift Card	1	60.06
	1	\$20.53	Completed	PayPal	1	20.53
	1	\$31.32	Completed	PayPal	1	31.32
	1	\$106.64	Completed	Debit Card	1	106.64
	2	\$34.82	Completed	Credit Card	2	34.82
	1	\$41.30	Completed	Gift Card	1	41.3
	1	\$130.75	Completed	Credit Card	1	130.75
	1	\$29.64	Completed		1	29.64
	1	\$16.59	completed	Debit Card	1	16.59
	1	\$38.05	Completed	Gift Card	1	38.05
	1	\$44.88	Completed	Credit Card	1	44.88
	2	\$12.32	Completed	Debit Card	2	12.32
	2	\$75.42	Completed	Debit Card	2	75.42
	3	\$15.48	cancelled	Gift Card	3	15.48

Gambar 13. Kolom unit_price_clean — tanda \$ berhasil dihapus, tersisa angka murni

Identifikasi dan penghapusan outlier

Setelah tipe data bersih, outlier diperiksa dengan mengurutkan (Sort Largest to Smallest) dan meninjau konteks tiap baris (produk, customer, kategori):

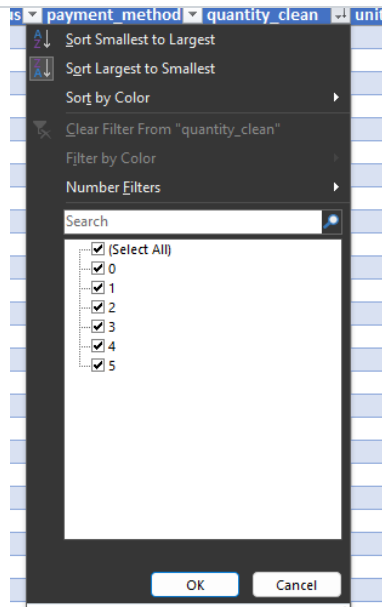
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	quantity_clean	unit_price	
1	ORD100335	2023-08-07	CUST11411	Ahmad Garcia	West	Home & Kitchen	Coffee Maker	999	43.04	Completed	PayPal	999	43.04	
2	ORD101178	2023-08-01	CUST11004	Ahmad Yusuf	South	Toys	Action Figure	999	35.43	completed	Gift Card	999	35.43	
3	ORD102638	2023-02-23	CUST11083	Ahmad Taylor	East	Sports & Outdoors	Yoga Mat	999	71.18	Completed	Gift Card	999	71.18	
4	ORD103218	2023-10-28	CUST11474	Kevin Sari	East	Beauty	Sunscreen SPF50	999	9.1	Completed	Gift Card	999	9.1	
5	ORD102875	2023-04-30	CUST11808	Michael Kurniawan	Midwest	Sports & Outdoors	Jump Rope	999	85.82	Completed	Credit Card	999	85.82	
6	ORD100684	2023-09-16	CUST11652	Ahmad Smith	West	Books	Business Strategy	999	17.71	Completed	Debit Card	999	17.71	
7	ORD102343	2023-07-25	CUST11772	Ahmad Hidayat		Electronics	Wireless Earbuds	500	52.43	Returned	Gift Card	500	52.43	
8	ORD103588	2023-01-01	CUST11104	Sarah Tan	South	Sports & Outdoors	Water Bottle	500	32.5	Cancelled	PayPal	500	32.5	
9	ORD100867	2023-01-31	CUST11399	Dewi Chen	West	Clothing	Winter Jacket	500	32.44	Completed	Gift Card	500	32.44	
10	ORD102002	2023-05-27	CUST11262	Robert Garcia	Midwest	Clothing	Hoodie	500	13.13	Completed	Credit Card	500	13.13	
11	ORD101689	2023-07-10	CUST11290	Lisa Nugroho	West	Home & Kitchen	Non-Stick Pan	500	98.01	Completed	Debit Card	500	98.01	
12	ORD100031	2024-05-22	CUST11179	David Salim	East	Sports & Outdoors	Water Bottle	250	9.42	Completed	Debit Card	250	9.42	
13	ORD100209	2024-01-20	CUST11761	Ahmad Johnson	East	Toys	Building Blocks	250	53.78	Completed		250	53.78	
14	ORD103734	2024-12-01	CUST11871	Andi Smith	West	Electronics	Laptop Stand	250	127.77	Cancelled	Debit Card	250	127.77	
15	ORD102727	2023-09-04	CUST11713	Dewi Taylor	West	Electronics	Webcam	250	44.7	Completed	Debit Card	250	44.7	
16	ORD102931	2023-10-26	CUST11856		South	Electronics	Power Bank	5	126.65	Completed	Gift Card	5	126.65	
17	ORD103343	2023-12-09	CUST11383	David Garcia	West	Sports & Outdoors	Hiking Backpack	5	68.67	Completed	Debit Card	5	68.67	
18	ORD100234	2023-12-22	CUST11766	Michael Kurniawan	South	Beauty	Sunscreen SPF50	5	37.89	Completed	PayPal	5	37.89	
19	ORD102815	2023-01-17	CUST11555	Andi Hidayat	Midwest	Toys	Action Figure	5	48.39	Completed	Debit Card	5	48.39	

Gambar 14. Kolom quantity_clean diurutkan dari terbesar — terlihat 15 baris dengan nilai 999/500/250

19	ORD100234	2024-09-25	CUST11716	Budi Salim	Midwest	Toys	Puzzle 1000pc	5	8888	Completed	Gift Card	5	8888	
20	ORD101663	2024-08-01	CUST11572	Maria Yusuf	East	Toys	Puzzle 1000pc	5	51.6	Cancelled	Debit Card	5	51.6	

Gambar 15. Salah satu baris outlier quantity (Puzzle 1000pc) kebetulan juga memiliki unit_price ekstrem (8888)

- Quantity: 15 baris dengan nilai 250/500/999 dihapus. Nilai tersebar acak di berbagai produk dan customer tanpa pola order borongan (wholesale) — diklasifikasikan sebagai data entry error, dan immaterial karena hanya 0,37% dari dataset.



Gambar 16. Filter kolom quantity setelah 15 baris outlier dihapus — tersisa nilai 0–5 saja

77	ORD101914	2023-09-23	CUST1838	Budi Chen	South	Home & Kitchen	Throw Pillow		Completed	PayPal		0
78	ORD100143	2024-10-24	CUST1799	Robert Hidayat	South	Toys	Building Blocks		Completed	Credit Card		0
79	ORD100533	2024-09-25	CUST1716	Budi Salim	Midwest	Toys	Puzzle 1000pc	5	8888 Completed	Gift Card	5	8888
80	ORD100693	2024-03-23	CUST1630	John Kurniawan	West	Home & Kitchen	Blender	1	8888 Returned	PayPal	1	8888
81	ORD101843	2024-12-16	CUST1023	Siti Wilson	East	Home & Kitchen	Non-Stick Pan	1	8888 Completed	Credit Card	1	8888
82	ORD102699	2024-01-20	CUST1454	Fajar Wijaya	Midwest	Clothing	Hoodie	1	8888 Completed	Gift Card	1	8888
83	ORD102755	2023-11-10	CUST1809	Kevin Smith	Midwest	Books	Business Strategy	2	4999.99 Completed	PayPal	2	4999.99
84	ORD101880	2024-05-06	CUST1535	Maria Brown	Midwest	Toys	Action Figure	1	4999.99 Cancelled	Credit Card	1	4999.99
85	ORD100921	2024-05-15	CUST1725	Nina Pratama	West	Books	Business Strategy	2	1234.56 Completed	Credit Card	2	1234.56
86	ORD101822	2024-11-14	CUST1564	Maria Chen	South	Sports & Outdoors	Jump Rope	2	1234.56 Completed	Credit Card	2	1234.56
87	ORD101110	2024-07-10	CUST1842	Siti Salim	East	Books	Cookbook	2	1234.56 Completed		2	1234.56
88	ORD101599	2024-12-06	CUST1138	Fajar Anderson	Midwest	Electronics	Power Bank	1	1234.56 Completed	Gift Card	1	1234.56
89	ORD101460	2024-01-01	CUST1145	Linda Smith	Midwest	Clothing	Wool Socks	1	1234.56 completed	Debit Card	1	1234.56
90	ORD100829	2023-02-16	CUST1399	Rina Lee	Midwest	Sports & Outdoors	Yoga Mat	1	1234.56 Completed	PayPal	0	1234.56
91	ORD101912	2024-07-21	CUST1634	James Sari	East	Electronics	Power Bank	2	149.71 Pending	Debit Card	2	149.71

Gambar 17. 12 baris dengan unit_price ekstrem (8888 / 4999,99 / 1234,56) diseleksi untuk dihapus

- Unit price: 12 baris dengan nilai 8888 / 4999,99 / 1234,56 dihapus. Nilai-nilai ini tidak sesuai kategori produknya (contoh: Puzzle 1000pc seharga \$8.888), dan nilai 1234,56 muncul identik berulang di 6 produk berbeda — indikasi kuat error sistem, bukan variasi harga wajar.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	quantity_clean	unit_price	
3939	ORD101018	2023-03-11	CUST1260	Fajar Pratama	East	Sports & Outdoors	Resistance Bands	1	Completed	Gift Card		1		
3939	ORD100917	2024-03-21	CUST1394	Maria Hidayat	South	Sports & Outdoors	Hiking Backpack	1	Returned	PayPal		1		
3939	ORD102799	2024-05-22	CUST1762	Rina Nugroho	South	Sports & Outdoors	Resistance Bands	1	Completed	Credit Card		1		
3939	ORD100944	2024-10-16	CUST1380	Lisa Brown	West	Beauty	Sunscreen SPF50	1	Completed	PayPal		1		
3939	ORD101530	2024-04-30	CUST1287	John Pratama	South	Sports & Outdoors	Yoga Mat	1	Pending	PayPal		1		
3939	ORD101879	2024-04-05	CUST1136	Budi Hidayat	Midwest	Books	Self-Help Guide	1	Completed	Debit Card		1		
3940	ORD103237	2023-10-11	CUST1624	Budi Smith	South	Home & Kitchen	Cutting Board	1	Completed	Gift Card		1		
3940	ORD100905	2024-05-15	CUST1350	Linda Salim	Midwest	Clothing	Hoodie	1	Completed	Credit Card		1		
3940	ORD103913	2023-01-24	CUST1026	Putri Yusuf	Midwest	Home & Kitchen	Cutting Board	1	Completed	Credit Card		1		
3940	ORD101125	2023-01-27	CUST1516	Fajar Brown	Midwest	Electronics	Webcam	1	Returned	Gift Card		1		
3940	ORD103870	2023-04-26	CUST1328	Andi Tan	South	Clothing	Denim Jeans	1	Returned	PayPal		1		
3940	ORD101487	2023-08-21	CUST1687	Siti Sari	South	Toys	Action Figure	1	Completed	Credit Card		1		
3940	ORD101406	2024-08-14	CUST1349	John Halim	East	Toys	Board Game	1	Completed	Gift Card		1		
3940	ORD102133	2024-11-30	CUST1745	Ahmad Halim	East	Clothing	Denim Jeans	1	Completed	Debit Card		1		
3940	ORD103818	2023-09-09	CUST1660	Andi Salim	East	Sports & Outdoors	Hiking Backpack	1	Completed	Credit Card		1		
3940	ORD100019	2024-08-06	CUST1412	Siti Anderson	South	Home & Kitchen	Coffee Maker	1	Completed	Debit Card		1		
3950	ORD101245	2024-04-24	CUST1412	Siti Anderson	West	Clothing	Wool Socks	1	Cancelled	PayPal		1		
3950	ORD103531	2024-02-27	CUST1894	Putri Anderson	East	Home & Kitchen	Blender	1	Completed	Gift Card		1		
3950	ORD101540	2024-06-01	CUST1754	Kevin Salim	West	Electronics	Power Bank	1	Returned	Debit Card		1		
3950	ORD100027	2023-05-05	CUST1324	Budi Santos	Midwest	Sports & Outdoors	Resistance Bands	1	Completed	Gift Card		1		
3950	ORD102052	2024-11-04	CUST1837	Devil Johnson	Midwest	Home & Kitchen	Blender	1	Completed	Debit Card		1		
3950	ORD103958	2024-03-30	CUST1005	Devil Wijaya	Toys	Building Blocks	1	Completed	Gift Card		1			
3950	ORD100155	2023-08-18	CUST1168	Linda Nugroho	East	Toys	Board Game	1	Completed	PayPal		1		
3950	ORD103247	2024-11-26	CUST1177	Ayu Pratama	East	Beauty	Sunscreen SPF50	1	Completed	Debit Card		1		
3950	ORD102778	2023-04-08	CUST1161	Ayu Yusuf	East	Books	Business Strategy	1	Completed	Credit Card		1		
3950	ORD103476	2024-10-03	CUST1171	James Anderson	Midwest	Books	Children's Book	1	Completed	PayPal		1		
3950	ORD100166	2024-06-30	CUST1846	Kevin Kurniawan	East	Books	Business Strategy	1	Returned	Gift Card		1		
3950	ORD100607	2023-07-01	CUST1617	Robert Tan	South	Beauty	Shampoo	1	Pending	Debit Card		1		
3960	ORD101989	2023-08-31	CUST1684	Fajar Salim	East	Toys	Puzzle 1000pc	1	Completed	PayPal		1		
3960	ORD103234	2024-09-18	CUST1642	South	Clothing	Hoodie	1	Returned	Gift Card		1			
3960	ORD101578	2024-11-30	CUST1256	James Wijaya	West	Clothing	Winter Jacket	1	Completed	Credit Card		1		
3960	ORD101526	2023-10-22	CUST1811	Putri Santos	Midwest	Clothing	Denim Jeans	1	Completed	PayPal		1		
3960	ORD100898	2023-05-02	CUST1575	West	Clothing	Denim Jeans one	Completed	Gift Card		1		1		
3960	ORD103206	2024-10-17	CUST1417	Electronics	Webcam	1	Completed	PayPal		1		1		
3960	ORD103339	2024-11-25	CUST1154	Michael Tan	East	Electronics	Laptop Stand	1	Completed	PayPal		1		
3960	ORD100759	2023-11-14	CUST1252	Ayu Brown	West	Home & Kitchen	Throw Pillow	1	Completed	Debit Card		1		
3970	ORD103614	2024-08-27	CUST1386	Lisa Johnson	Midwest	Books	Business Strategy	1	Completed	Credit Card		1		
3970	ORD102400	2024-07-10	CUST1324	Budi Santos	Midwest	Toys	Action Figure	1	Completed	Debit Card		1		
3970	ORD101194	2024-07-13	CUST1705	Ayu Brown	Midwest	Electronics	Smartphone Case	Completed	Credit Card		0			
3970	ORD100216	2024-09-12	CUST1356	Ahmad Tan	Midwest	Home & Kitchen	Throw Pillow	Pending	Gift Card		0			
3970	ORD101914	2023-09-23	CUST1838	Budi Chen	South	Home & Kitchen	Throw Pillow	Completed	PayPal		0			
3970	ORD100143	2024-10-24	CUST1799	Robert Hidayat	South	Toys	Building Blocks	Completed	Credit Card		0			

Gambar 18. Sebelum: outlier quantity/unit_price masih tercampur di dalam data

#	A	B	C	D	E	F	G	H	I	J	K	L	M
1	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	quantity_clean	unit_price_clean
7	ORD100006	2024-01-16	CUST1580	Linda Tan	South	Electronics	Smartphone Case	1	81.85	Completed	Credit Card	1	81.85
8	ORD100007	2023-09-26	CUST1753	Kevin Lee	South	Clothing	Running Shorts	1	77.87	Completed	Gift Card	1	77.87
9	ORD100008	2024-11-29	CUST1075	Linda Anderson	South	Sports & Outdoors	Water Bottle	1	43.76	Pending	Debit Card	1	43.76
10	ORD100009	2023-05-05	CUST1726	John Tan	Midwest	Beauty	Shampoo	1	36.12	Pending	PayPal	1	36.12
11	ORD100010	2024-02-12	CUST1132	Robert Tan	East	Books	Fiction Novel	1	10.00	Pending	Gift Card	1	10
12	ORD100011	2024-07-18	CUST1460	Rina Sari	Midwest	Sports & Outdoors	Jump Rope	3	36.61	Completed	PayPal	3	36.61
13	ORD100012	2023-12-21	CUST1311	John Nugroho	Clothing	Denim jeans	3	18.37	Returned	Debit Card	3	18.37	
14	ORD100013	2024-11-05	CUST1546	John Garcia	West	Electronics	Smartphone Case	1	72.72	Completed	Debit Card	1	72.72
15	ORD100014	1900-01-00	CUST1590	Putri Tan	West	Home & Kitchen	LED Desk Lamp	3	93.60	Completed	Gift Card	3	93.6
16	ORD100015	2023-06-11	CUST1515	Sarah Salim	East	Home & Kitchen	Cutting Board	4	106.12	Completed	Gift Card	4	106.12
17	ORD100016	2023-11-23	CUST1353	Andi Halim	Sports & Outdoors	Water Bottle	2	28.97	Completed	Credit Card	2	28.97	
18	ORD100017	2023-07-14	CUST1254	Nina Garcia	Midwest	Toys	Puzzle 1000pc	1	25.48	Completed	Gift Card	1	25.48
19	ORD100018	2023-02-21	CUST1468	Michael Tan	Midwest	Clothing	Denim jeans	1	35.43	Completed	Gift Card	1	35.43
20	ORD100019	2024-08-06	CUST1412	Siti Anderson	South	Home & Kitchen	Coffee Maker	1		Completed	Debit Card	1	
21	ORD100020	2024-02-08	CUST1696	David Hidayat	West	Books	Fiction Novel	1	28.56	Completed	Debit Card	1	28.56
22	ORD100021	2024-06-15	CUST1039	Robert Tan	Midwest	Clothing	Wool Socks	1	54.35	Completed	PayPal	1	54.35
23	ORD100022	2024-01-05	CUST1344	Ahmad Salim	South	Books	Business Strategy	1	14.12	Completed	Debit Card	1	14.12
24	ORD100023	2023-02-22	CUST1644	Andi Salim	Midwest	Electronics	Wireless Mouse	2	67.21	Completed	PayPal	2	67.21
25	ORD100024	2024-03-27	CUST1586	Putri Smith	West	Beauty	Moisturizer	1	9.63	Completed	PayPal	1	9.63
26	ORD100025	2024-11-09	CUST1744	Andi Sari	West	Electronics	Smartphone Case	3	100.47	Completed	Gift Card	3	100.47
27	ORD100026	2023-09-10	CUST1738	Fajar Halim	South	Beauty	Face Mask Set	2	9.96	Completed	Debit Card	2	9.96
28	ORD100027	2023-05-05	CUST1324	Budi Santos	Midwest	Sports & Outdoors	Resistance Bands	1		Completed	Gift Card	1	
29	ORD100028	2024-05-05	CUST1731	Robert Anderson	South	Electronics	Laptop Stand	1	115.78	Completed	Debit Card	1	115.78
30	ORD100029	2023-04-13	CUST1685	Budi Kurniawan	West	Books	Cookbook	1	25.04	Completed	Gift Card	1	25.04
31	ORD100030	2023-10-22	CUST1416		South	Beauty	Lip Balm	1	31.17	Completed	Credit Card	1	31.17
32	ORD100032	2024-05-31	CUST1656	Ayu Yusuf	South	Books	Business Strategy	4	16.56	Completed		4	16.56
33	ORD100033	2024-09-17	CUST1345	Michael Santos	West	Toys	RC Car	4	35.43	Cancelled	Gift Card	4	35.43
34	ORD100034	2023-08-30	CUST1331	Ayu Chen	Midwest	Clothing	Cotton T-Shirt	3	58.79	Completed	Debit Card	3	58.79
35	ORD100035	2023-06-13	CUST1063	David Anderson	South	Beauty	Moisturizer	3	33.38	Completed	Debit Card	3	33.38
36	ORD100036	2024-12-19	CUST1489	Nina Salim	Midwest	Clothing	Running Shorts	3	36.29	Completed	Gift Card	3	36.29
37	ORD100037	1900-01-00	CUST1806	Ahmad Hidayat	Midwest	Electronics	Wireless Mouse	1	94.76	Completed	PayPal	1	94.76
38	ORD100038	2024-04-09	CUST1597	Ahmad Nugroho	South	Sports & Outdoors	Yoga Mat	2	12.75	Completed	Gift Card	2	12.75
39	ORD100039	2024-04-09	CUST1836	Kevin Salim	East	Electronics	USB-C Cable	2	146.74	Completed	Credit Card	2	146.74
40	ORD100040	2024-01-27	CUST1184	Michael Kurniawan	South	Toys	Puzzle 1000pc	1	21.76	Completed	Gift Card	1	21.76
41	ORD100041	2023-01-04	CUST1182	Dewi Tan	West	Books	Children's Book	0	27.92	Completed	Debit Card	0	27.92
42	ORD100042	2024-07-11	CUST1158	Nina Nugroho	East	Home & Kitchen	Non-Stick Pan	1	93.85	Completed	Credit Card	1	93.85
43	ORD100043	2023-08-28	CUST1250	Linda Smith	South	Beauty	Moisturizer	3	12.75	Completed	Credit Card	3	12.75
44	ORD100044	2023-08-27	CUST1391	Fajar Wilson	West	Books	Self-Help Guide	1	24.90	Pending	PayPal	1	24.9
45	ORD100045	2023-12-19	CUST1695	James Johnson	South	Sports & Outdoors	Hiking Backpack	2	10.44	Returned	Gift Card	2	10.44
46	ORD100046	2024-12-25	CUST1484	James Anderson	West	Clothing	Baseball Cap	1	42.08	Completed	Gift Card	1	42.08

Gambar 19. Sesudah: kolom quantity dan unit_price bersih, tanpa nilai ekstrem

Hari	Fase	Task	Tools	Selesai?
Day 1	Excel Data Cleaning	Ambil dataset e-commerce sales	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek & tangani missing values	Excel	
Day 1	Excel Data Cleaning	Cek & hapus duplicate records	Excel	SELESAI
Day 1	Excel Data Cleaning	Perbaiki data types yang salah	Excel	SELESAI
Day 1	Excel Data Cleaning	Standarkan kategori yang tidak konsisten	Excel	SELESAI
Day 1	Excel Data Cleaning	Format ulang kolom tanggal	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek outlier	Excel	SELESAI
Day 1	Excel Data Cleaning	Buat calculated columns	Excel	
Day 1	Excel Data Cleaning	Dokumentasikan raw -> cleaning -> clean dataset	Excel	
Day 1	Excel Data Cleaning	Simpan sebagai cleaned_sales_data.xlsx	Excel	

Gambar 20. Checklist Day 1 setelah tahap perbaikan tipe data dan outlier selesai

3.5 Penanganan Missing Value

Setelah tahap sebelumnya, enam kolom masih memiliki missing value dengan jumlah dan tingkat kekritisannya berbeda-beda. Jumlah missing value per kolom dihitung menggunakan COUNTBLANK di area kosong di luar tabel:

```
=COUNTBLANK(D2:D3975) {ganti huruf kolom sesuai field yang dicek}
```

SUM	:	X	✓	f_x	=COUNTBLANK(D2:D3975)
1	order_id	order_date	customer_id	customer_name	region
3963	ORD103988	2024-04-15	CUST1412	Siti Anderson	West
3964	ORD103989	2023-06-18	CUST1696	David Hidayat	West
3965	ORD103990	2024-12-01	CUST1774	Maria Wijaya	West
3966	ORD103991	2023-11-10	CUST1664	Dewi Nugroho	Midwest
3967	ORD103992	2023-04-25	CUST1292	Budi Halim	South
3968	ORD103993	2024-03-03	CUST1382	David Hidayat	Midwest
3969	ORD103994	2024-01-17	CUST1058	John Johnson	West
3970	ORD103995	2023-02-16	CUST1113	Dewi Nugroho	Midwest
3971	ORD103996	2023-02-05	CUST1248	Andi Pratama	East
3972	ORD103997	2024-02-22	CUST1494	Robert Pratama	East
3973	ORD103998	2023-11-05	CUST1348	Putri Nugroho	East
3974	ORD103999	2023-01-20	CUST1421	Ayu Lee	South
3975	ORD104000	2024-02-12	CUST1580	Linda Tan	Midwest
3976					
3977					
3978					
3979					
3980				customer_name	
3981				D3975)	
3982					
3983					
3984					
3985					

Gambar 21. Formula COUNTBLANK diterapkan di luar tabel untuk menghitung missing value tiap kolom

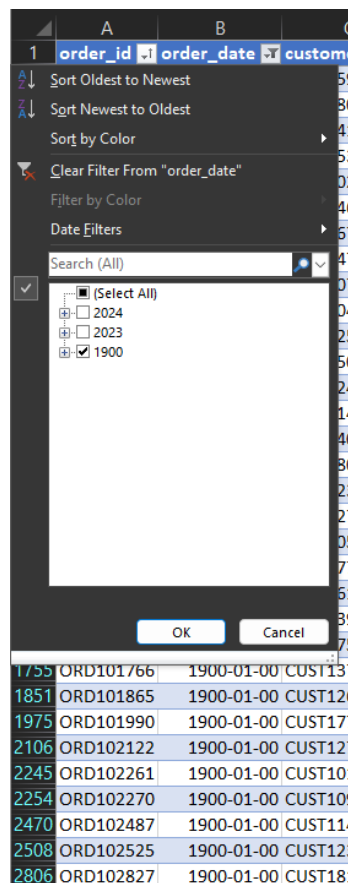
customer_name		120
region		79
payment_method		158
order_date		0
quantity		0
unit_price		77

Gambar 22. Hasil rekap awal — order_date sempat terbaca 0 karena nilai kosong tersembunyi sebagai angka 0

customer_name		120
region		79
payment_method		158
order_date		0
quantity		59
unit_price		77

Gambar 23. Rekap setelah quantity yang sempat tersamar sebagai 0 dikembalikan menjadi blank (naik menjadi 59)

order_date sempat terbaca 0 missing, padahal ada nilai yang tersembunyi sebagai angka 0 (bekas sel kosong yang diformat jadi tanggal 1 Januari 1900). Ditemukan lewat filter tahun pada kolom order_date:



Gambar 24. Filter order_date menemukan baris dengan tahun 1900 — bekas sel kosong yang tersembunyi

customer_name	120
region	79
payment_method	158
order_date	40
quantity	59
unit_price	77

Gambar 25. Rekap final missing value per kolom setelah semua nilai tersembunyi ditemukan

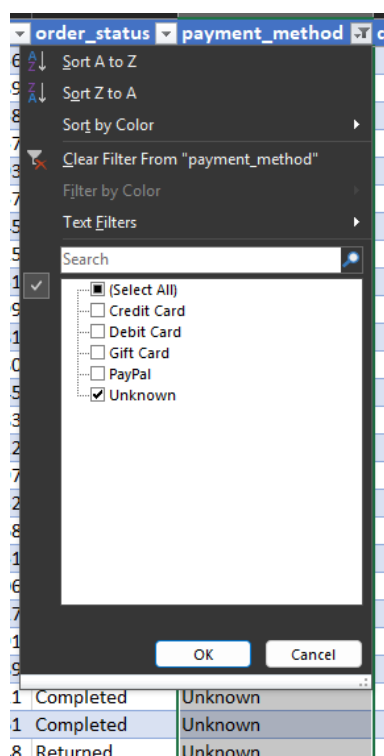
Strategi penanganan disesuaikan per kolom — kolom kategorikal diisi label "Unknown", kolom yang dipakai untuk perhitungan (quantity, unit_price) diisi menggunakan estimasi statistik, sedangkan order_date dihapus karena tidak dapat direkonstruksi secara valid.

Kolom	Jumlah	%	Strategi Penanganan
payment_method	158	4,0%	Diisi "Unknown" — kolom kategori, tidak memengaruhi perhitungan.
customer_name	120	3,0%	Diisi "Unknown Customer" — identitas tetap tersedia via customer_id.
region	79	2,0%	Direcovery dari transaksi lain milik customer_id yang sama (INDEX/MATCH array formula); sisanya diisi "Unknown".
unit_price	77	1,9%	Diisi rata-rata harga per product_name (AVERAGEIFS) — lebih akurat dibanding rata-rata global.
quantity	59	1,5%	Diisi median quantity keseluruhan (1 unit) — median dipilih karena data condong ke angka kecil.
order_date	40	1,0%	Baris dihapus — tanggal transaksi krusial untuk analisis tren dan tidak dapat direkonstruksi secara valid.

Tabel 2. Ringkasan missing value dan strategi penanganannya

payment_method dan customer_name

Kedua kolom kategorikal ini diisi memakai filter (Blanks) lalu Select All + ketik nilai pengganti + Alt+Enter untuk mengisi seluruh sel yang terseleksi sekaligus:



Gambar 26. Filter payment_method menampilkan baris kosong, siap diisi "Unknown"

region — dicoba direcovery lebih dulu

Kolom region dicoba direcovery dari transaksi lain milik customer_id yang sama sebelum jatuh ke "Unknown", menggunakan formula array (ditekan dengan Ctrl+Shift+Enter):

```
=IFERROR(INDEX($region$2:$region$3975, MATCH(1,
($customer_id$2:$customer_id$3975=[@customer_id]) * ($region$2:$region$3975<>""), 0)),
"") {Ctrl+Shift+Enter}
```

	A	B	C	D	E	F	G	H	I	J	K	L
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	region_recovered
2	ORD100553	2024-11-04	CUST1305	Linda Nugroho		Home & Kitchen	Storage Container	1	31.12	Completed	Debit Card	East
3	ORD101005	2023-04-05	CUST1755	Budi Sari		Electronics	Smartphone Case	1	29.75	Returned	PayPal	East
4	ORD101116	2023-11-02	CUST1733	Lisa Sari		Beauty	Face Mask Set	1	29.15	Cancelled	Debit Card	East
5	ORD101125	2023-01-27	CUST1516	Fajar Brown		Electronics	Webcam	1		Returned	Gift Card	East
6	ORD101190	2023-03-17	CUST1638	Maria Kurniawan		Toys	Building Blocks	1	29.17	Completed	Credit Card	East
7	ORD101319	2023-01-24	CUST1707	Michael Yusuf		Electronics	Power Bank	3	70.28	Completed	Debit Card	East
8	ORD101435	2023-05-15	CUST1844	Ahmad Johnson		Clothing	Baseball Cap	5	59.21	Completed	PayPal	East
9	ORD101524	2024-01-18	CUST1172	Nina Lee		Home & Kitchen	Blender	1	78.32	Completed	Gift Card	East
10	ORD101662	2024-04-08	CUST1121	David Nugroho		Home & Kitchen	Storage Container	1	24.34	Completed	PayPal	East
11	ORD102055	2024-08-04	CUST1708	Ayu Smith		Beauty	Face Mask Set	2	32.63	Completed	Gift Card	East
12	ORD102156	2024-01-11	CUST1798	Siti Anderson		Beauty	Lip Balm	1	6.38	Completed	Gift Card	East
13	ORD102170	2023-10-16	CUST1561	Nina Santos		Toys	Building Blocks	2	41.55	Completed	Debit Card	East
14	ORD102589	2024-05-25	CUST1558	Sarah Wilson		Sports & Outdoors	Hiking Backpack	2	56.86	Completed	Credit Card	East
15	ORD102759	2024-12-22	CUST1572	Maria Yusuf		Toys	Board Game	2	52.88	Cancelled	Credit Card	East
16	ORD102912	2024-03-14	CUST1851	Robert Salim		Toys	Action Figure	4	33.78	Completed	Gift Card	East
17	ORD103206	2024-10-17	CUST1417	Unknown Customer		Electronics	Webcam	1		Completed	PayPal	East
18	ORD103366	2024-11-14	CUST1723	Ayu Wijaya		Home & Kitchen	LED Desk Lamp	2	43.94	Completed	Debit Card	East
19	ORD103436	2023-07-04	CUST1402	Rina Nugroho		Clothing	Wool Socks	1	33.39	Completed	PayPal	East

Gambar 27. Kolom region_recovered berhasil menemukan region dari transaksi lain customer yang sama

unit_price dan quantity — diisi estimasi statistik

Kolom unit_price diisi memakai AVERAGEIFS — rata-rata harga dihitung khusus dari produk yang sama (product_name), bukan rata-rata seluruh dataset:

```
=IF([@unit_price]="", AVERAGEIFS($unit_price$2:$unit_price$3975,
$product_name$2:$product_name$3975, [@product_name]), [@unit_price])
```

unit_price	order_status	payment_method	unit_price_fill
	Completed	Credit Card	83.09465909
	Pending	Gift Card	63.9183908
	Completed	Credit Card	36.80701923
	Completed	PayPal	63.9183908

Gambar 28. Kolom unit_price_fill — harga kosong terisi rata-rata harga produk yang sama

Kolom quantity diisi memakai MEDIAN (bukan AVERAGE), karena data quantity condong ke angka kecil (1–5) sehingga median lebih tahan terhadap distorsi dibanding rata-rata:

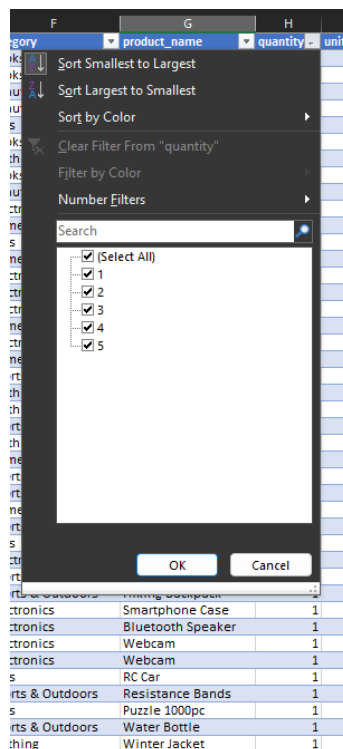
```
=IF([@quantity]="", MEDIAN($quantity$2:$quantity$3975), [@quantity])
```

quantity	unit_price	order_status	payment_method	unit_price_fill	quantity_fill
21.69	Completed	Credit Card		21.69	1
11.03	Completed	Gift Card		11.03	1
58.91	Completed	Credit Card		58.91	1
122.70	Completed	Debit Card		122.7	1
17.39	Completed	Credit Card		17.39	1
49.25	Returned	Credit Card		49.25	1
13.42	Completed	Debit Card		13.42	1
64.16	Completed	Unknown		64.16	1
33.37	Completed	Gift Card		33.37	1
58.85	Completed	Gift Card		58.85	1
19.42	Completed	Debit Card		19.42	1
76.05	Completed	Debit Card		76.05	1
75.06	Completed	Debit Card		75.06	1
	Completed	Credit Card		83.09465909	1
17.66	Completed	Debit Card		17.66	1

Gambar 29. Kolom unit_price_fill dan quantity_fill tampil bersama — quantity kosong konsisten terisi median (1)

Catatan cepat — sintaks dasar ketiga fungsi ini:

- **AVERAGE(range)** — rata-rata semua angka di range, tanpa syarat.
- **AVERAGEIFS(average_range, criteria_range1, criteria1, ...)** — rata-rata dari average_range, tapi hanya baris yang memenuhi syarat (bisa lebih dari satu syarat).
- **MEDIAN(range)** — nilai tengah dari sekumpulan angka; lebih tahan terhadap outlier dibanding AVERAGE.



Gambar 30. Filter quantity setelah semua missing value tertangani — hanya tersisa nilai wajar 1–5

Menggabungkan hasil kembali ke kolom asli

Saat memindahkan hasil kolom bantu (seperti `region_recovered`) kembali ke kolom asli, cara yang lebih efisien untuk baris dalam jumlah banyak adalah mengurutkan (Sort A-Z) kolom bantu terlebih dahulu — baris yang berhasil direcovery akan berkumpul jadi satu grup, sehingga bisa disalin sekaligus per grup, bukan satu per satu:

	A	B	C	D	E	F	G	H	I	J	K	L	M
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	unit_price_fill	quantity_fill
29	ORD103119		CUST1171	James Anderson	East	Beauty	Shampoo	1	8.19	Completed	PayPal	8.19	1
333	ORD101366		CUST1407	Rina Smith	East	Books	Children's Book	2	13.80	Returned	PayPal	13.80	2
586	ORD100368		CUST1472	Andi Halim	Midwest	Beauty	Shampoo	2	17.57	Completed	Gift Card	17.57	2
65	ORD103834		CUST1853	Robert Smith	East	Beauty	Face Mask Set	2	18.12	Completed	Credit Card	18.12	2
830	ORD102827		CUST1818	Siti Salim	East	Toys	RC Car	1	21.00	Completed	Credit Card	21.00	1
845	ORD102261		CUST1016	John Pratama	Midwest	Beauty	Lip Balm	1	21.12	Completed	Debit Card	21.12	1
947	ORD101212		CUST1503	Andi Anderson	East	Beauty	Sunscreen SPF50	1	22.65	Returned	Gift Card	22.65	1
987	ORD100583		CUST1076	Kevin Santos	East	Clothing	Running Shorts	2	23.24	Completed	PayPal	23.24	2
1058	ORD100121		CUST1538	Dewi Kurniawan	East	Toys	RC Car	1	24.15	Returned	Gift Card	24.15	1
1074	ORD100824		CUST1040	Lisa Garcia	East	Toys	Puzzle 1000pc	2	24.38	Cancelled	PayPal	24.38	2
1088	ORD101865		CUST1287	Maria Anderson	East	Home & Kitchen	Blender	2	24.78	Completed	PayPal	24.78	2
1378	ORD100265		CUST1023	Siti Wilson	East	Home & Kitchen	Non-Stick Pan	1	28.43	Completed	Debit Card	28.43	1
1500	ORD101196		CUST1254	Nina Garcia	East	Beauty	Sunscreen SPF50	2	30.21	returned	Gift Card	30.21	2
1649	ORD101990		CUST1173	Andi Nugroho	Midwest	Sports & Outdoors	Resistance Bands	2	32.18	Completed	Debit Card	32.18	2
1670	ORD101307		CUST1248	Andi Pratama	East	Toys	Board Game	1	32.42	Completed	Credit Card	32.42	1
1833	ORD100317		CUST1674	Dewi Lee	East	Sports & Outdoors	Resistance Bands	1	35.10	Completed	PayPal	35.10	1
1940	ORD102270		CUST1093	Kevin Nugroho	Midwest	Electronics	Smartphone Case	1	37.47	Completed	PayPal	37.47	1
2167	ORD101408		CUST1808	Michael Kurniawan	Midwest	Home & Kitchen	Coffee Maker	1	41.66	Pending	Credit Card	41.66	1
2283	ORD103651		CUST1376	Nina Yusuf	Midwest	Toys	Action Figure	2	44.14	Completed	Gift Card	44.14	2
2300	ORD101765		CUST1757	Sarah Lee	East	Beauty	Moisturizer	1	44.45	Cancelled	Debit Card	44.45	1
2677	ORD103515		CUST1178	Kevin Wilson	East	Home & Kitchen	Throw Pillow	2	54.89	completed	Gift Card	54.89	2
2707	ORD100294		CUST1462	Michael Lee	Midwest	Toys	RC Car	4	55.58	Completed	Credit Card	55.58	4
2917	ORD101628		CUST1050	Sarah Anderson	East	Sports & Outdoors	Jump Rope	2	61.56	Completed	Debit Card	61.56	2
2934	ORD103310		CUST1374	Nina Garcia	Midwest	Clothing	Cotton T-Shirt	1	62.48	Completed	Credit Card	62.48	1
3108	ORD101786		CUST1379	Putri Halim	East	Sports & Outdoors	Resistance Bands	2	69.99	Completed	Debit Card	69.99	2
3113	ORD101704		CUST1198	Nina Kurniawan	East	Electronics	Smartphone Case	1	70.07	Completed	PayPal	70.07	1
3120	ORD101325		CUST1149	Sarah Yusuf	Midwest	Clothing	Cotton T-Shirt	1	70.18	Completed	Gift Card	70.18	1
3167	ORD102122		CUST1178	David Salim	East	Sports & Outdoors	Camping Tent	2	72.10	Completed	PayPal	72.10	2
3260	ORD101426		CUST1278	David Salim	East	Electronics	USB-C Cable	1	76.10	Pending	PayPal	76.10	1
3301	ORD101644		CUST1174	Maria Wijaya	East	Home & Kitchen	Non-Stick Pan	1	77.96	completed	Credit Card	77.96	1
3377	ORD100056		CUST1415	Sarah Salim	East	Sports & Outdoors	Hiking Backpack	1	82.78	Completed	Credit Card	82.78	1
3516	ORD101650		CUST1618	Putri Wijaya	Midwest	Electronics	Power Bank	1	91.79	Completed	PayPal	91.79	1
3551	ORD100014		CUST1590	Putri Tan	East	Home & Kitchen	LED Desk Lamp	3	93.60	Completed	Gift Card	93.60	3
3573	ORD100037		CUST1806	Ahmad Hidayat	East	Electronics	Wireless Mouse	1	94.76	Completed	PayPal	94.76	1
3618	ORD101425		CUST1238	John Pratama	East	Electronics	Wireless Earbuds	2	97.63	Completed	Debit Card	97.63	2
3706	ORD102525		CUST1233	Michael Lee	East	Electronics	Smartphone Case	2	104.81	Returned	Debit Card	104.81	2
3734	ORD102969		CUST1816	John Kurniawan	East	Home & Kitchen	Coffee Maker	1	107.95	Completed	Gift Card	107.95	1
3738	ORD103137		CUST1086	Rina Anderson	West	Home & Kitchen	Non-Stick Pan	2	108.13	Pending	PayPal	108.13	2
3778	ORD102975		CUST1246	Ayu Brown	East	Electronics	Laptop Stand	1	112.19	Completed	Debit Card	112.19	1
3868	ORD102487		CUST1143	Linda Wijaya	East	Electronics	Laptop Stand	1	129.86	Completed	PayPal	129.86	1
3976													

Gambar 31. Kolom *region_recovered* diurutkan A-Z untuk mempermudah penyalinan hasil per grup ke kolom asli

- *order_date*: 40 baris kosong dihapus karena tanggal transaksi krusial untuk analisis tren dan tidak bisa direkonstruksi secara valid.

3.6 Pembuatan Calculated Column (Revenue)

Kolom revenue ditambahkan sebagai perkalian sederhana antara *quantity* dan *unit_price*, sehingga nilainya otomatis konsisten dengan seluruh proses cleaning sebelumnya:

```
=[@quantity]*[@unit_price]
```

H	I	J	K	L	M
quantity	unit_price	order_status	payment_method	quantity_fill	revenue
1	6.00	Completed	Gift Card	1	6.00
1	6.22	Completed	Gift Card	1	6.22
2	6.22	Completed	Credit Card	2	12.44
2	6.27	Completed	Debit Card	2	12.54
1	6.38	Completed	Gift Card	1	6.38
4	6.40	Completed	Debit Card	4	25.60
2	6.72	Completed	Gift Card	2	13.44
1	6.80	Returned	PayPal	1	6.80
2	6.84	Returned	PayPal	2	13.68
2	6.86	Completed	Credit Card	2	13.72

Gambar 32. Kolom *revenue* pada baris-baris awal setelah formula diterapkan

H	I	J	K	L	M	N	O
quantity	unit_price	order_status	payment_method	quantity_fill	revenue		
1	6.00	Completed	Gift Card	1	6.00		
1	6.22	Completed	Gift Card	1	6.22		
2	6.22	Completed	Credit Card	2	12.44		
2	6.27	Completed	Debit Card	2	12.54		
1	6.38	Completed	Gift Card	1	6.38		
4	6.40	Completed	Debit Card	4	25.60		
2	6.72	Completed	Gift Card	2	13.44		
1	6.80	Returned	PayPal	1	6.80		
2	6.84	Returned	PayPal	2	13.68		
2	6.86	Completed	Credit Card	2	13.72		
1	6.92	Completed	PayPal	1	6.92		
2	6.95	Completed	Debit Card	2	13.90		
1	6.96	Completed	Debit Card	1	6.96		
1	6.98	Completed	Gift Card	1	6.98		
1	6.98	Completed	Gift Card	1	6.98		
1	7.06	Pending	Credit Card	1	7.06		
2	7.18	Completed	PayPal	2	14.36		
4	7.19	Completed	Gift Card	4	28.76		
1	7.25	Completed	Gift Card	1	7.25		
1	7.26	Completed	Gift Card	1	7.26		
4	7.44	Completed	PayPal	4	29.76		
2	7.83	Completed	PayPal	2	15.66		
1	7.84	Completed	Gift Card	1	7.84		
1	7.88	Completed	Debit Card	1	7.88		
5	7.95	Completed	Credit Card	5	39.75		
1	8.13	Pending	Debit Card	1	8.13		
2	8.17	Completed	Credit Card	2	16.34		
3	8.20	Completed	Debit Card	3	24.60		
2	8.22	Completed	Debit Card	2	16.44		
3	8.23	Cancelled	Gift Card	3	24.69		
1	8.29	Completed	Gift Card	1	8.29		
1	8.29	Completed	Debit Card	1	8.29		
1	8.33	Completed	Gift Card	1	8.33		
1	8.39	Completed	Credit Card	1	8.39		
1	8.41	Completed	Gift Card	1	8.41		
1	8.41	Completed	Credit Card	1	8.41		
1	8.41	Pending	Credit Card	1	8.41		
1	8.43	Completed	Debit Card	1	8.43		
3	8.47	Returned	Credit Card	3	25.41		
1	8.49	Completed	Gift Card	1	8.49		
1	8.56	Pending	Gift Card	1	8.56		
1	8.59	Completed	PayPal	1	8.59		
3	8.64	Completed	PayPal	3	25.92		
1	8.75	Completed	Debit Card	1	8.75		
1	8.75	Completed	Gift Card	1	8.75		
1	8.87	Completed	Credit Card	1	8.87		
1	8.88	Completed	Gift Card	1	8.88		
1	8.90	Completed	PayPal	1	8.90		
1	8.91	Pending	Credit Card	1	8.91		
2	8.97	Completed	Gift Card	2	17.94		
2	8.98	Completed	Gift Card	2	17.96		
1	9.04	Completed	Gift Card	1	9.04		
1	9.04	Completed	Credit Card	1	9.04		
2	9.04	Completed	Debit Card	2	18.08		
1	9.05	Completed	Credit Card	1	9.05		
4	9.05	Completed	PayPal	4	36.20		
1	9.05	Completed	Unknown	1	9.05		
1	9.06	Completed	Debit Card	1	9.06		
1	9.06	Completed	PayPal	1	9.06		
3	9.07	Completed	Debit Card	3	27.21		
1	9.08	Completed	PayPal	1	9.08		

Gambar 33. Kolom *revenue* pada dataset — total revenue \$306.888,66 dari 3.934 baris (sebelum 1 duplikat tambahan ditemukan pada verifikasi ulang di bagian 3.1)

Hari	Fase	Task	Tools	Selesai?
Day 1	Excel Data Cleaning	Ambil dataset e-commerce sales	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek & tangani missing values	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek & hapus duplicate records	Excel	SELESAI
Day 1	Excel Data Cleaning	Perbaiki data types yang salah	Excel	SELESAI
Day 1	Excel Data Cleaning	Standarkan kategori yang tidak konsisten	Excel	SELESAI
Day 1	Excel Data Cleaning	Format ulang kolom tanggal	Excel	SELESAI
Day 1	Excel Data Cleaning	Cek outlier	Excel	SELESAI
Day 1	Excel Data Cleaning	Buat calculated columns	Excel	SELESAI
Day 1	Excel Data Cleaning	Dokumentasikan raw -> cleaning -> clean dataset	Excel	
Day 1	Excel Data Cleaning	Simpan sebagai cleaned_sales_data.xlsx	Excel	

Gambar 34. Checklist Day 1 — seluruh tahap cleaning selesai, tinggal dokumentasi dan penyimpanan file

4. Ringkasan Hasil (Cleaning Log)

Tabel berikut merangkum seluruh langkah cleaning secara berurutan, dan disertakan sebagai sheet terpisah ("Cleaning Log") dalam file cleaned_sales_data.xlsx sebagai jejak audit proses.

Langkah	Baris Terdampak	Keterangan
1. Hapus duplicate records	99 baris	Exact duplicate (seluruh kolom identik), diduga input ganda — 98 baris pada pembersihan awal + 1 baris ditemukan saat verifikasi ulang di Day 2 (teknik row signature + filter warna).
2. Standardisasi category & region	—	Diseragamkan ke Proper Case; typo diperbaiki (Cloths→Clothing, Electronic→Electronics, Mid-West→Midwest).
3. Standardisasi order_date	—	5 format tanggal disatukan menjadi YYYY-MM-DD.
4. Hapus outlier quantity	15 baris	Nilai ekstrem (250/500/999) tidak masuk akal untuk transaksi retail individual, tersebar random tanpa pola wholesale.
5. Hapus outlier unit_price	12 baris	Nilai ekstrem (8888/4999,99/1234,56) tidak sesuai kategori produk; nilai identik berulang mengindikasikan error sistem.
6. Tangani missing value	533 sel	Lihat Tabel 2 — strategi berbeda per kolom sesuai tingkat kepentingan & kemungkinan recovery.
7. Buat calculated column revenue	3.933 baris	revenue = quantity × unit_price. Total revenue dataset: \$306.888,66.

Tabel 3. Cleaning Log — ringkasan seluruh proses raw dataset → clean dataset

5. Kesimpulan

Dataset akhir berisi 3.933 baris bersih, tanpa missing value, dengan tipe data dan format yang konsisten pada seluruh kolom. Total 167 baris (4,1% dari data awal) dihapus sepanjang proses, seluruhnya melalui keputusan yang terdokumentasi dan dapat dipertanggungjawabkan, bukan penghapusan sepihak tanpa alasan. Satu duplikat tambahan yang lolos dari pembersihan awal berhasil ditemukan lewat verifikasi ulang sebelum data dipakai di tahap analisis — praktik quality assurance yang penting dilakukan sebelum data dianggap final.

Output tahap ini adalah file cleaned_sales_data.xlsx yang berisi dua sheet: data transaksi yang sudah bersih, dan Cleaning Log sebagai dokumentasi proses. Dataset ini menjadi sumber tunggal yang dipakai secara konsisten pada tahap-tahap berikutnya: analisis Excel (Day 2), query SQL (Day 3), exploratory data analysis dengan Python (Day 4), dan dashboard Tableau (Day 5).